

Topic 9: Mixing and separating materials

Students should:		Guidance <i>iPrimary Curriculum reference</i>
9.1	understand that solids can be mixed and that sieving may be used to separate some mixtures	<i>Year 5: Mixing and separating materials</i>
9.2	explain how filtration may be used to separate some solids from a liquid	a simple explanation only: - liquids pass through filter paper - insoluble solids do not. the terms <i>filtrate</i> and <i>residue</i> are NOT required recall sand and flour are examples of insoluble solids <i>Year 5: Mixing and separating materials</i>
9.3	understand that when a solid dissolves in water it forms a solution that cannot be separated by filtration	recall sugar and salt are examples of soluble solids <i>Year 5: Mixing and separating materials</i>
9.4	know ways in which simple substances such as sugar and salt can be dissolved more quickly	including: - crushing the solid - heating the liquid - stirring the mixture. interpreting data in simple tables or graphs to compare the amounts of different solids that will dissolve in a given volume of water <i>Year 5: Mixing and separating materials</i>
9.5	explain that when a solution is left exposed to the air the liquid will evaporate into the air leaving the dissolved solid behind	<i>Year 5: Mixing and separating materials</i>
9.6	use knowledge of solids, liquids and gases to decide how mixtures might be separated, including by sieving, using a magnet, filtering and evaporating	use knowledge about how a specific mixture can be separated to suggest ways in which other similar mixtures may be separated <i>Year 5: Mixing and separating materials</i>